

Notes: Cookson et al. (JFE, 2026)

Social media as a bank run catalyst

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1 Big picture

- **Question.** Does **social media** amplify classical **bank run risk**, beyond the traditional balance-sheet vulnerabilities that already make a bank fragile?
- **Setting.** The paper studies the March 2023 run on **Silicon Valley Bank (SVB)** and the broader distress among U.S. regional banks, combining **comprehensive Twitter data**, bank balance-sheet data, stock-market data, and deposit-flow data.
- **Core challenge.** It is hard to tell whether Twitter is merely reflecting bad fundamentals or actually amplifying a run. Banks with more attention on social media may also be larger, more salient, or already viewed as risky. The paper therefore needs to separate **pre-existing exposure to social media** from **classical run risk**, and then study timing carefully.
- **Main contribution.** The paper combines three layers of evidence:
 1. cross-sectional tests that relate **Twitter pre-exposure** before the run to **run-period stock losses**;
 2. cross-sectional tests that relate Twitter pre-exposure to **Q1-2023 deposit outflows**;
 3. high-frequency hourly and tweet-level tests that distinguish the role of **Twitter attention** from **Twitter sentiment**.
- **Why it matters.** Modern bank runs can unfold through digital banking and rapid online communication. If social media is an important coordination device, then traditional models of bank fragility remain right in spirit, but the speed and propagation mechanism are different.
- **Main empirical question.** Are banks that were more exposed to Twitter before the SVB collapse more vulnerable during the run, especially when they also had large **uninsured deposits** and large **mark-to-market losses**?
- **Main empirical results.**
 - A one-standard-deviation increase in Twitter pre-exposure predicts about **4.3 percentage points** more stock-market loss during the SVB run.
 - Twitter pre-exposure **interacts positively** with classical run-risk variables—especially % uninsured deposits and mark-to-market losses—to predict more severe stock declines.
 - The same interaction also predicts larger **Q1-2023 deposit outflows**, especially for **uninsured deposits**.
 - At the hourly frequency, high recent Twitter attention predicts more negative returns **only for high-run-risk banks**; for low-run-risk banks, the same relation is absent.
 - High-frequency tweet-level tests show that **run-relevant tweets**—especially those by the **tech community** and tweets about **running** or **contagion**—move bank stock prices immediately.
 - By contrast, **negative Twitter sentiment does not amplify bank run risk**. The paper's core mechanism is about **attention and coordination**, not simply negativity.
- **Economic takeaway.** Social media appears to act as a **bank run catalyst**: it does not replace classical balance-sheet risk, but it can amplify it by increasing attention, communication, and likely coordination among depositors.

2 Data and measurement (key objects)

2.1 Twitter data and text measurement

- **Main source.** The authors use academic-access Twitter API data.
- **Main sample window.** The raw Twitter collection spans **January 1, 2020 to March 14, 2023**.
- **Tweet sample.** The main sample keeps **original English tweets** containing a bank cashtag (e.g. \$SIVB) for depository institutions with SIC codes 602, 603, or 609.
- **Sample size.**
 - **5,399,740** original tweets about banking firms from 2020 through March 2023.
 - **765,224** retweets from January 1 to March 31, 2023 in a separate retweet sample.
 - **544,888** unique Twitter users who authored at least one sample tweet.
- **Tweet-level variables.** The data include tweet text, timestamp, author identifier, and retweet counts.
- **Author-level variables.** The paper collects user profile descriptions, account age, and follower counts.
- **Sentiment.** The paper scores each tweet using **VADER** (Valence Aware Dictionary and sEntiment Reasoner), yielding a score on $[-1, 1]$ and separate positive and negative components.
- **Context dictionaries.** The authors build dictionaries for:
 - **balance-sheet** tweets,
 - **run-behavior** tweets,
 - **contagion** tweets,
 - and **tech community** users (based on profile words such as venture capital, founder, entrepreneur, startup).
- **Why the tech-community dictionary matters.** A key claim of the paper is that likely SVB depositors were highly concentrated in the tech/startup ecosystem, which is also highly active on Twitter.

2.2 Bank sample, balance sheets, stock prices, and deposits

- **Bank universe for the main cross-section.** The Twitter sample initially covers **359** banks with nonzero pre-period Twitter attention. Merging with intraday stock data leaves **335** banks; merging with balance-sheet risk variables leaves a final cross-sectional sample of **277** publicly traded banks and bank holding companies.
- **Stock price data.** Intraday stock prices come from **FirstRate Data**. These are used to measure:
 - run-period stock losses,

- hourly returns during the SVB run,
- and minute-level returns around tweet timestamps.
- **Run-period loss measure.** The main return outcome is the percentage stock-value loss from **March 1 to March 14, 2023**.
- **Deposit-flow data.** Deposit outflows are measured using **FDIC Call Reports**, comparing Q4-2022 to Q1-2023 deposits, separately for **uninsured deposits** and **total deposits**.
- **Uninsured deposits.** The variable **% Uninsured** is computed from deposits above and below the FDIC insurance threshold of \$250,000.
- **Mark-to-market balance-sheet risk.** The paper constructs a market-based measure of unrealized asset losses by mapping banks' maturity exposures into Treasury-index price changes over 2022, with an adjustment for mortgage-backed securities.
- **Core run-risk measure.** The paper defines

$$\text{RunRisk}_i = \% \text{Uninsured}_i \times \% \text{LossMTM}_i.$$

This is the paper's summary measure of balance-sheet fragility.

2.3 Pre-run and run-period definitions

- **Pre-run period.** January 1, 2023 through February 15, 2023.
- **Run-period tweet window.** Most run-period tweet counts are built over roughly **March 8 through March 13**.
- **Hourly panel window.** March 6 through March 10, 2023, with the run onset defined at **March 9 at 9 a.m.**
- **Tweet-level event window.** For each tweet, the return is measured from the last trade just before the tweet to the first trade in the **5-to-15 minute** window after the tweet.

2.4 Key descriptive facts

- **Twitter attention rises sharply during the run.** On a 30-day equivalent basis, banks average about **537** tweets in the pre-run period and about **2,279** during the run period.
- **Twitter attention is persistent.** The correlation between pre-run tweet counts and in-run tweet counts is about **0.85**. This is central to the interpretation of pre-exposure as a proxy for future attention capacity.
- **Pre-exposure is not just hidden run risk.** Twitter pre-exposure is only weakly correlated with pre-period tweet sentiment (about $r = 0.18$) and essentially uncorrelated with run risk (about $r = 0.02$).
- **SVB dominates the run conversation.** During the run period, SVB receives roughly **6,528** *run* tweets and **9,662** *contagion* tweets, far above the rest of the sample.

- **Investor tweets lead general discussion.** Tweets using the ticker **SIVB** spike before generic mentions of **SVB** or **Silicon Valley Bank**, suggesting that investor/depositor discussion led broader public discussion.
- **Tech-community tweets matter.** Tweets by accounts associated with founders/startups/VCs spike distinctly during the run and are more likely to mention the bank directly rather than the stock ticker, consistent with actual depositor coordination.

2.5 Unobservables and timing

- **Cross-sectional concern.** Banks with more Twitter attention may also differ in size, media coverage, analyst following, valuation, deposit franchise, or prior returns.
- **High-frequency concern.** Tweets during the run may react to price moves rather than cause them.
- **Paper’s solution.** The design uses:
 - **pre-run** Twitter exposure for the cross-section,
 - **lagged** Twitter attention in the hourly panel,
 - and **narrow windows around tweet timestamps** for the minute-level tests.

3 Models and empirical specifications (all equations + interpretation)

3.1 Mark-to-market loss construction and run risk

The paper first builds a bank-level proxy for unrealized losses from duration exposure:

$$\%LossMTM_i \approx \sum_m AssetExposure_{im} \times \Delta TreasuryPrice_m,$$

where m indexes maturity or repricing buckets, with an additional adjustment for mortgage-backed securities.

It then defines the main balance-sheet risk variable as

$$RunRisk_i = \%Uninsured_i \times \%LossMTM_i.$$

Interpretation. The paper’s core idea is that a bank is especially fragile when it combines a large uninsured deposit base with large unrealized asset losses. The social-media question is whether Twitter amplifies the effect of this classical fragility measure.

3.2 Determinants of Twitter pre-exposure

To study what predicts social-media exposure before the run, the paper estimates:

$$TwitterPreExposure_i = a + b_1 Size_i + b_2 \%LossMTM_i + b_3 \%Uninsured_i + b_4 X_i + b_5 W_i + b_6 Z_i + \varepsilon_i.$$

Here:

- $\text{TwitterPreExposure}_i$ is the log number of tweets about bank i in the pre-run period,
- X_i collects information-environment controls such as analyst coverage and news coverage,
- W_i collects market controls such as prior returns, FOMC-event returns, and Google search attention,
- Z_i collects deposit and balance-sheet controls.

Interpretation. This is not the main causal equation. Instead, it is a credibility exercise. The authors show that Twitter pre-exposure is strongly related to bank **size** and to some general salience variables, but only weakly related to the fundamental run-risk variables. That supports using pre-exposure as a measure of social-media exposure rather than latent fragility.

3.3 Cross-sectional regression for run-period stock losses

The main stock-loss specification is

$$\text{StockLoss}_i = \beta_1 (\% \text{LossMTM}_i \times \% \text{Uninsured}_i \times \text{PreExp}_i) + \text{lower-order terms} + \gamma X_i + \varepsilon_i.$$

Where:

- StockLoss_i is the percentage stock-value loss during the SVB run,
- PreExp_i is Twitter pre-exposure,
- X_i includes size, valuation, attention, prior stock returns, and deposit-franchise controls.

Interpretation. The triple interaction is the key coefficient. A positive coefficient means that social-media exposure **amplifies classical bank-run risk**: banks with both high uninsured deposits and large mark-to-market losses suffer bigger stock losses when they also had more Twitter pre-exposure.

3.4 Cross-sectional regression for deposit outflows

The deposit-flow specification mirrors the stock-loss specification:

$$\text{DepositOutflow}_i = \beta_1 (\% \text{LossMTM}_i \times \% \text{Uninsured}_i \times \text{PreExp}_i) + \text{lower-order terms} + \gamma X_i + \varepsilon_i.$$

The dependent variable is the percentage loss of deposits from Q4-2022 to Q1-2023, estimated separately for **uninsured deposits** and **total deposits**.

Interpretation. If the same interaction that predicts stock-market losses also predicts actual deposit outflows, then stock-price evidence is more plausibly capturing run severity rather than unrelated equity-market news.

3.5 Hourly panel regression for lagged Twitter attention

For bank i in hour t , the paper estimates

$$\begin{aligned} r_{it} = & a + b_1 1(t \geq \text{Mar } 9, 9\text{am}) + b_2 1(\text{TweetsHigh}_{i,t-1}) \\ & + b_3 1(t \geq \text{Mar } 9, 9\text{am}) \times 1(\text{TweetsHigh}_{i,t-1}) + \eta X_{i,t-1} + \delta_i + \gamma_t + \varepsilon_{it}. \end{aligned}$$

Here:

- r_{it} is the hourly return in basis points,
- $1(\text{TweetsHigh}_{i,t-1})$ indicates whether tweet volume in the prior 4 hours is above the median,
- δ_i are bank fixed effects,
- γ_t are day-by-hour fixed effects,
- $X_{i,t-1}$ includes lagged stock returns and lagged news coverage.

Interpretation. This regression uses **lagged attention** to address timing. The key interaction asks whether recent Twitter attention predicts worse returns *after* the run begins, and whether this only happens for high-run-risk banks.

3.6 Tweet-level high-frequency return regression

Around each tweet timestamp, the paper estimates

$$\Delta p_{it} = a + b \text{VADERPos}_{it} + c \text{VADERNeg}_{it} + \eta X_{it} + \gamma_i + \varepsilon_{it},$$

with additional interactions between sentiment and indicators for:

- tweets authored by **tech community** users,
- **run** tweets,
- **contagion** tweets,
- and banks with **high run risk**.

The dependent variable Δp_{it} is the change in log price from the last trade just before the tweet to the first trade in the 5-to-15 minute window after the tweet.

Interpretation. These tests isolate the immediate price effect of tweet content. They are especially useful for asking whether negative sentiment itself drives price reactions, or whether the economically important force is tweet **attention and relevance**.

3.7 Attention versus sentiment

The empirical design makes a sharp distinction:

- **Attention** is measured by tweet counts or high/low tweet-activity indicators.
- **Sentiment** is measured by VADER positive and negative components.

Interpretation. This distinction is one of the main conceptual contributions of the paper. The authors show that **attention** amplifies bank run risks, whereas **sentiment** does not.

4 Identification and instruments (what makes the estimates credible)

4.1 What would be unconvincing in a simpler approach

- A simple correlation between tweets and losses would be weak evidence because tweets might just reflect public news or falling prices.

- Likewise, heavily discussed banks may simply be larger, more followed, or more vulnerable *ex ante*.
- During the run itself, contemporaneous tweet counts could be endogenous to returns.

4.2 What identifies the cross-sectional evidence

- **Pre-period design.** Twitter exposure is measured *before* the run, over January 1 to February 15, 2023.
- **Content validation.** The pre-run period contains little direct discussion of bank distress; the paper emphasizes that pre-period attention is not driven by bank-run talk.
- **Persistence in attention.** Pre-run exposure strongly predicts in-run attention (correlation about 0.85), making it a meaningful proxy for a bank's capacity to become focal on Twitter once a run starts.
- **Weak relation to run fundamentals.** Pre-exposure is almost uncorrelated with the paper's run-risk measure and only weakly related to pre-run tweet sentiment.
- **Rich control set.** The main regressions control for bank size, analyst coverage, news coverage, prior stock returns, deposit beta, branch footprint, market-to-book ratio, deposit flows, CRE exposure, liquid assets, and deposit concentration.

4.3 What identifies the high-frequency evidence

- **Lagged Twitter attention.** In the hourly regressions, tweet activity is measured over the previous 1, 2, or 4 hours.
- **Narrow event windows.** In the tweet-level tests, returns are measured only in the few minutes after a tweet.
- **Controls for recent prices and news.** The tweet-level regressions condition on stock-price changes and RavenPack news measures from the 10 minutes before the tweet.
- **Fixed effects.** Bank fixed effects absorb time-invariant differences across banks; day-by-hour fixed effects absorb common run dynamics.

4.4 Why the results point to coordination rather than generic negativity

- If Twitter were just a reflection of bad news, one would expect **negative sentiment** to be the key force.
- Instead, the hourly results show that **attention** matters for high-run-risk banks, while sentiment does not amplify run risk.
- The immediate price effects are strongest for **run-relevant tweets** and **tech-community tweets**, which is exactly what one would expect if depositor communication is the mechanism.
- The effect is stronger when tweets are **retweeted heavily**, reinforcing the idea that the social propagation of attention matters.

4.5 Robustness and external credibility

- The results survive using alternative pre-periods going back to October 2022.
- They remain similar when excluding **too-big-to-fail** banks.
- They remain similar when excluding **SVB** from the hourly sample, though magnitudes weaken somewhat.
- The balance-sheet loss measure is checked against fair-value disclosures in 10-K filings.
- A specification-curve exercise shows that the triple interaction between pre-exposure and run risk remains positive across a large number of specifications.

4.6 Relation to the literature

- **Bank run theory.** The paper is closest in spirit to Diamond–Dybvig style fragility and Goldstein–Pauzner style coordination models.
- **2023 banking-stress literature.** It complements recent work emphasizing uninsured deposits, mark-to-market losses, commercial real estate exposure, and flights to safety.
- **Social media and finance.** It extends the financial-social-media literature from asset-pricing effects to a setting where online communication can help drive a real financial crisis mechanism.

5 Tables and figures

5.1 Table 1, Figure 1, and Figure 2: dictionaries and the anatomy of the SVB conversation

Table 1
Contextual dictionaries for classifying types of tweets.

Balance sheet	Run behavior	Contagion	Tech community
dividend	run	systemic	VC
cover & cash	withdraw	adviser	entrepreneur
margin-backed securities	deposit money	fed	start up
mismatch	access accounts	regulator	startup
long maturity	pull it out	frustration	founder
maturity mismatch		backlog	venture
market to market		whole system	venture capital
crack in market		spreads	
portfolio management		leader effects	
liquidity		financial system	
insured deposits		collateral	
MSB		congestion	
hold to maturity			
portfolio of loans			
liquidity management			
uninsured deposits			
balance sheet			

This table presents the keyword dictionaries for the major content dictionaries that we employ to classify tweets in to “Run,” “Contagion,” “Balance sheet,” as well as “Tech Community” authors. For each of our content dictionaries we provide the most search used, in the investor creation process, in italics.

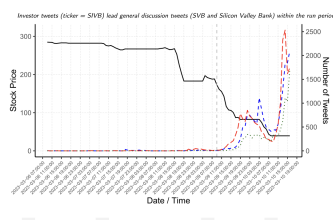


Fig. 1. Silicon Valley Bank stock price and Twitter mentions. This figure shows the stock price of Silicon Valley Bank (SVB) (black solid line), as well as the number of tweets mentioning “SVB” (blue dashed line), “SVB” (red solid line), and “Silicon Valley Bank” (green dotted line), over the period from March 6, 2023 to the trader close to March 15, 2023 at 16:00 (see trader close). The stock price is indicated on the left y-axis and the number of tweets is indicated on the right y-axis. The grey vertical dashed line indicates March 9, 2023 at 16:00.

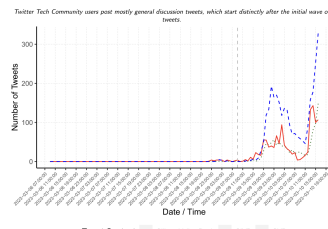


Fig. 2. Tech Community Tweets about Silicon Valley Bank. This figure shows the number of tweets mentioning “SVB” (blue solid line), “SVB” (red solid line), and “Silicon Valley Bank” (green dotted line), over the period from March 6, 2023 at 16:00 (trader close) to March 15, 2023 at 16:00 (trader close). The sample includes only tweets posted by users who are part of the tech community as defined by corresponding dictionary of terms in the Twitter user biography. The grey vertical dashed line indicates March 9, 2023 at 16:00.

- **Read.** Table 1 defines the main dictionaries for balance-sheet, run, contagion, and tech-community content.
- **Figure 1.** Investor-centric ticker tweets about **SVB** spike before broader mentions of **SVB** or **Silicon Valley Bank**. This is important because it suggests the initial conversation came from financially engaged users rather than from generic news chatter.
- **Figure 2.** Tech-community tweets spike prominently during the run, and they focus directly on the bank. This supports the interpretation that likely depositors were active in the online conversation.

- **Punchline.** The socially relevant Twitter conversation is not random noise. It is tightly tied to the bank-run narrative and to the likely depositor base.

5.2 Figures 3–6 and Table 3: pre-exposure construction and validation

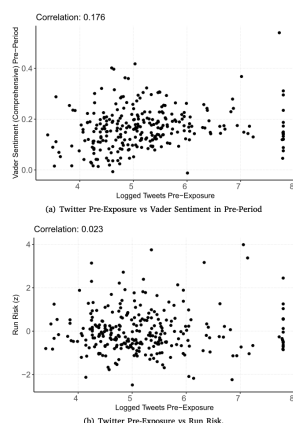
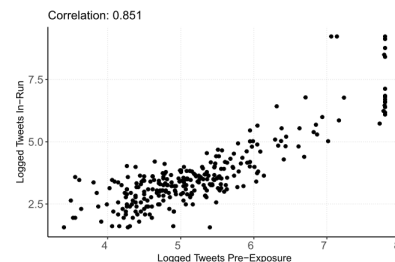
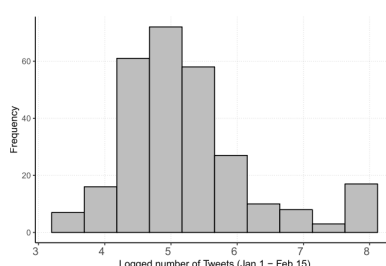
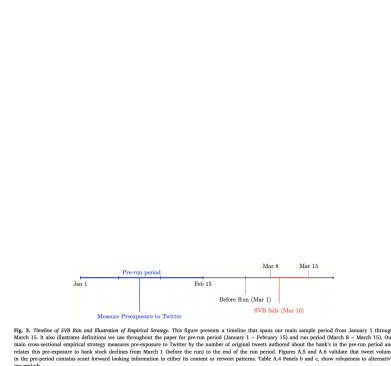


Table 3
Determinants of twitter pre-exposure.

	Social media pre-exposure				
	(1)	(2)	(3)	(4)	(5)
ln(MCap) (z)	0.767*** (0.043)	0.730*** (0.047)	0.738*** (0.054)	0.744*** (0.050)	0.691*** (0.074)
% Loss MTM (z)	-0.027 (0.040)	-0.025 (0.039)	-0.026 (0.039)	-0.066* (0.037)	-0.085** (0.037)
% Uninsured (z)	-0.051 (0.050)	-0.056 (0.051)	-0.055 (0.051)	-0.074 (0.046)	-0.067 (0.046)
Log N Articles (z)	0.101** (0.043)	0.111** (0.058)	0.087 (0.053)	0.045 (0.054)	0.037 (0.052)
Log N Analysts (z)		-0.019 (0.071)	-0.054 (0.072)	0.004 (0.075)	0.015 (0.070)
Ret 2022-Q4 (z)			-0.010 (0.042)	-0.009 (0.041)	0.009 (0.040)
Ret 2022-Q3 (z)			-0.078* (0.040)	-0.075** (0.038)	-0.064* (0.038)
FOMC 2022 CAR (z)			-0.183*** (0.050)	-0.170*** (0.045)	-0.125*** (0.047)
4 SVI Q4-Q3 (z)			0.052 (0.047)	0.043 (0.049)	0.047 (0.046)
% Dep Flow 2022-Q4 (z)				-0.042 (0.053)	0.103 (0.085)
N ZIPs Branches (z)				0.064 (0.044)	0.063 (0.043)
Deposit Beta (z)				-0.025 (0.038)	-0.035 (0.037)
MTB BHC (z)				-0.083** (0.034)	-0.036 (0.034)
% CRE Loans (z)					-0.141*** (0.041)
% Liquid Assets (z)					0.061 (0.058)
Deposit Concentration (z)					0.071*** (0.019)
Constant	5.235*** (0.038)	5.235*** (0.038)	5.235*** (0.038)	5.234*** (0.036)	5.234*** (0.035)
Observations	277	277	277	277	274
R ²	0.588	0.597	0.597	0.643	0.654
Adjusted R ²	0.583	0.591	0.590	0.631	0.637

This table presents OLS regressions that relate Twitter Pre-Exposure to characteristics of the bank, market and information environment that could drive attention to banks. The dependent variable is 'Twitter pre-exposure', i.e., the log of the number of tweets about a given bank in the pre-run period from Jan 1 through Feb 15, 2023. The explanatory variables include bank characteristics such as size (ln(MCap)) and valuation (MTB), run risk variables (% Loss MTM and % Uninsured depositors), attention characteristics (analyst coverage and number of newspaper articles in 2022, change in Google Search Volume SVI between Q4 and Q3 of 2022), stock returns (in Q3 and Q4 of 2022 and around 2022 FOMC meetings), measures of the deposit franchise (i.e., deposit beta from Drechsler et al. (2017) and the number of unique states with bank branches), and measures of deposit characteristics (% of commercial-real-estate (CRE) loans, deposit flow in Q4-2022, supply-driven deposit inflows in 2022, and deposit concentration). All variables denoted with '(z)' are standardized to have a mean of zero and standard deviation of one to ease magnitude comparisons.

* Indicate statistical significance at the 10% level.

** Indicate statistical significance at the 5% level.

*** Indicate statistical significance at the 1% level.

- **Read.** Figure 3 lays out the timing of the empirical strategy: pre-exposure is measured before the run, then related to run-period outcomes.
- **Figure 4.** The distribution of pre-exposure is well behaved once winsorized.
- **Figure 5.** Pre-run and in-run tweet activity are very tightly related (correlation about **0.85**).
- **Figure 6.** Pre-exposure is only weakly related to pre-period tweet sentiment ($r \approx 0.18$) and essentially unrelated to run risk ($r \approx 0.02$).
- **Table 3.** The strongest predictor of pre-exposure is **size**; traditional run-risk variables are weakly related to it. This is crucial for the paper's identification argument.

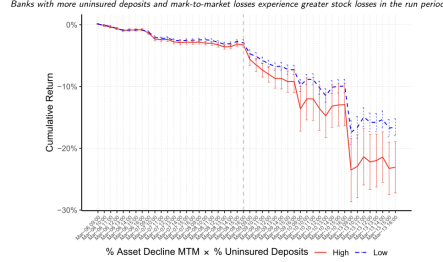


Fig. 7. *Cumulative Returns of Banks with High versus Low Run Risk.* This figure shows the average cumulative returns of US publicly listed banks and bank holding companies (BHCs) from March 06 through March 13, 2023. The sample is split by “Run Risk” defined as the % of uninsured deposits $\times$ the % MTM-implied asset value decline. “High” represents the top half above the median (red solid line). “Low” represents the bottom half (blue dashed line). The gray vertical dashed line indicates March 09, 2023 at 9am. 90% confidence intervals around the mean cumulative return are constructed by bootstrapping with clustering at the bank level.

5.3 Figure 7 and Table 4: stock-market evidence on amplification

Table 4
Twitter pre-exposure and bank stock losses

Panel A: Baseline model

	(1)	(2)	(3)
% Stock Value lost during run			
Pre-Exposure (z) $\times$ % Uninsured (z) $\times$ % Loss MTM (z)	1.437*** (0.266)	1.447*** (0.264)	1.447*** (0.264)
Pre-Exposure (z) $\times$ % Loss MTM (z)	1.382*** (0.328)	1.441*** (0.313)	1.441*** (0.313)
Pre-Exposure (z)	1.389*** (0.363)	1.447*** (0.342)	1.447*** (0.342)
% Uninsured (z) $\times$ % Loss MTM (z)	1.794*** (0.724)	1.822*** (0.744)	1.822*** (0.744)
% Uninsured (z)	1.845*** (0.448)	1.845*** (0.448)	1.845*** (0.448)
% Loss MTM (z)	0.665 (0.665)	0.665 (0.665)	0.665 (0.665)
% Loss MTM (z) $\times$ % Loss MTM (z)	2.407*** (0.865)	2.407*** (0.865)	2.407*** (0.865)
Constant	1.657 (1.044)	1.308 (1.079)	1.308 (1.079)
Log % Assets (z)	1.647*** (0.749)	1.921*** (0.761)	1.921*** (0.761)
Log % Assets (z)	0.264 (0.355)	-0.522 (0.616)	-0.469 (0.520)
Run 2022-Q4 (z)	-0.059*** (0.327)	-0.128*** (0.449)	-0.128*** (0.449)
Run 2022-Q3 (z)	-0.004 (0.705)	-0.246 (0.705)	-0.246 (0.705)
FORC 2022 CAR (z)	-0.363 (0.512)	-0.363 (0.512)	-0.363 (0.512)
Adj R ² Q4-Q3 (z)	-0.696 (0.512)	-0.671 (0.512)	-0.671 (0.512)
% Day Flow 2022-Q4 (z)	-0.420*** (0.797)	-0.420*** (0.797)	-0.420*** (0.797)
% Day Flow 2022-Q4 (z)	0.325 (0.632)	0.325 (0.632)	0.325 (0.632)
Deposit Run (z)	0.888 (0.357)	0.888 (0.357)	0.888 (0.357)
MTM Run (z)	1.532*** (0.502)	1.532*** (0.502)	1.532*** (0.502)
% CDR Loans (z)	-0.327 (0.449)	-0.327 (0.449)	-0.327 (0.449)
% Liquid Assets (z)	0.786 (0.786)	0.786 (0.786)	0.786 (0.786)
Deposit Concentration (z)	0.812 (0.463)	0.812 (0.463)	0.812 (0.463)
Constant	16.550*** (2.77)	16.113*** (2.77)	16.113*** (2.77)
Observations	277	277	277
Adjusted R ²	0.141	0.142	0.143

(continued on next page)

Table 4 (continued)

Panel B: Including controls for size, valuation, attention, prior stock returns, deposit flows, deposit franchise, risk

	(1)	(2)	(3)	(4)
% Stock Value lost during run				
Pre-Exposure (z) $\times$ % Uninsured (z) $\times$ % Loss MTM (z)	1.437*** (0.266)	1.447*** (0.264)	1.447*** (0.264)	1.447*** (0.264)
Pre-Exposure (z) $\times$ % Loss MTM (z)	1.382*** (0.328)	1.441*** (0.313)	1.441*** (0.313)	1.441*** (0.313)
Pre-Exposure (z)	1.389*** (0.363)	1.447*** (0.342)	1.447*** (0.342)	1.447*** (0.342)
% Uninsured (z) $\times$ % Loss MTM (z)	1.794*** (0.724)	1.822*** (0.744)	1.822*** (0.744)	1.822*** (0.744)
% Uninsured (z)	1.845*** (0.448)	1.845*** (0.448)	1.845*** (0.448)	1.845*** (0.448)
% Loss MTM (z)	0.665 (0.665)	0.665 (0.665)	0.665 (0.665)	0.665 (0.665)
% Loss MTM (z) $\times$ % Loss MTM (z)	2.407*** (0.865)	2.407*** (0.865)	2.407*** (0.865)	2.407*** (0.865)
Log % Assets (z)	1.647*** (0.749)	1.921*** (0.761)	1.921*** (0.761)	1.921*** (0.761)
Log % Assets (z)	0.264 (0.355)	-0.522 (0.616)	-0.469 (0.520)	-0.469 (0.520)
Run 2022-Q4 (z)	-0.059*** (0.327)	-0.128*** (0.449)	-0.128*** (0.449)	-0.128*** (0.449)
Run 2022-Q3 (z)	-0.004 (0.705)	-0.246 (0.705)	-0.246 (0.705)	-0.246 (0.705)
FORC 2022 CAR (z)	-0.363 (0.512)	-0.363 (0.512)	-0.363 (0.512)	-0.363 (0.512)
Adj R ² Q4-Q3 (z)	-0.696 (0.512)	-0.671 (0.512)	-0.671 (0.512)	-0.671 (0.512)
% Day Flow 2022-Q4 (z)	-0.420*** (0.797)	-0.420*** (0.797)	-0.420*** (0.797)	-0.420*** (0.797)
% Day Flow 2022-Q4 (z)	0.325 (0.632)	0.325 (0.632)	0.325 (0.632)	0.325 (0.632)
Deposit Run (z)	0.888 (0.357)	0.888 (0.357)	0.888 (0.357)	0.888 (0.357)
MTM Run (z)	1.532*** (0.502)	1.532*** (0.502)	1.532*** (0.502)	1.532*** (0.502)
% CDR Loans (z)	-0.327 (0.449)	-0.327 (0.449)	-0.327 (0.449)	-0.327 (0.449)
% Liquid Assets (z)	0.786 (0.786)	0.786 (0.786)	0.786 (0.786)	0.786 (0.786)
Deposit Concentration (z)	0.812 (0.463)	0.812 (0.463)	0.812 (0.463)	0.812 (0.463)
Constant	16.550*** (2.77)	16.113*** (2.77)	16.113*** (2.77)	16.113*** (2.77)
Observations	277	277	277	277
Adjusted R ²	0.141	0.142	0.143	0.144

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Table 4 (continued)

Panel C: Controlling for recent activity during run period

	(1)	(2)	(3)	(4)	(5)
% Stock Value lost during run					
Pre-Exposure (z)	4.317*** (0.558)	0.421 (0.558)	-0.408 (0.549)	-1.276 (1.110)	-2.862 (0.671)
Contagion Tweets In-Run (z)					3.942*** (1.460)
Tech-Community Tweets In-Run (z)					3.988*** (1.362)
Tech-Community Tweets In-Run (z)					3.988*** (1.362)
Constant	16.113*** (2.497)	16.090*** (2.497)	16.113*** (2.497)	16.113*** (2.497)	16.113*** (2.497)
Observations	277	277	277	277	277
Adjusted R ²	0.185	0.185	0.185	0.185	0.185
Funding controls					
Market controls					
Deposits controls					
Attention controls					
Valuation controls					

This table presents OLS estimates of regressions that relate bank stock losses during the run period to Twitter pre-exposure, mark-to-market losses, and percentage of uninsured deposits (Panel A presents the baseline results). All variables derived with “z” are standardized to have mean zero and standard deviation of one. Panel B is similar to Panel A but additionally controls for bank size and valuation, attention characteristics, stock returns, measures of the deposit franchise and measures of deposit characteristics. Variable definitions are as in Table 1, with full detail on their construction in Appendix Table A.1. In Panel C, we add “Contagion” tweets during the run (the number of tweets with at least one word in the “Run” tag) and “Tech-Community” tweets during the run (the number of tweets with at least one word in the “Tech” tag) to the set of controls. Robust standard errors are reported in parentheses.

*** Indicate statistical significance at the 1% level.

- **Figure 7.** Banks with high run risk suffer much larger cumulative stock losses, and the divergence begins at the onset of the SVB run on **March 9**.
- **Table 4, Panel A.** Without controls, a one-standard-deviation increase in Twitter pre-exposure predicts about **4.317 percentage points** more stock-market loss.
- **Table 4, Panel A.** The triple interaction between pre-exposure, uninsured deposits, and mark-to-market losses is about **2.407 percentage points** without controls.
- **Table 4, Panel B.** With controls, the triple interaction remains large and significant, about **1.44 to 1.66 percentage points** depending on specification.
- **Table 4, Panel C.** Once the regressions control for in-run **run**, **contagion**, and **tech-community** tweet counts, the coefficient on pre-exposure shrinks sharply. This supports the mechanism that pre-exposure matters because it generates more run-relevant attention during the crisis.

5.4 Table 5: deposit-flow evidence

- **Read.** Table 5 checks whether the same logic appears in actual deposit data.
- **Uninsured outflows.** The interaction between run risk and pre-exposure is positive and significant, with the triple interaction around **1.47** without controls and **1.78** with controls.

Table 5
Twitter pre-exposure and deposit flows.

	Deposit outflows Q1-2023 (%)			
	Uninsured		Total	
	(1)	(2)	(3)	(4)
% Uninsured (z) × % Loss MTM (z) × Pre-Exposure (z)	1.47** (0.746)	1.78** (0.890)	0.673 (0.536)	1.51*** (0.476)
% Uninsured (z) × Pre-Exposure (z)	1.59 (0.993)	0.989 (1.03)	1.74** (0.697)	0.897 (0.547)
% Loss MTM (z) × Pre-Exposure (z)	1.79* (0.923)	0.208 (1.19)	0.788 (0.704)	0.501 (0.679)
Pre-Exposure (z)	0.973 (1.12)	2.03 (1.29)	0.974 (0.968)	0.512 (0.839)
% Uninsured (z) × % Loss MTM (z)	-0.878 (0.652)	-1.24 (0.835)	0.112 (0.369)	-0.260 (0.459)
% Uninsured (z)	3.31*** (1.17)	4.25*** (1.42)	1.37*** (0.502)	1.83*** (0.542)
% Loss MTM (z)	0.609 (0.759)	0.081 (0.886)	0.829* (0.470)	0.088 (0.545)
ln(MCap) (z) × % Uninsured (z) × % Loss MTM (z)		0.683 (0.896)		0.201 (0.580)
ln(MCap) (z) × % Uninsured (z)		2.23* (1.35)		1.89** (0.728)
ln(MCap) (z) × % Loss MTM (z)		2.68* (1.55)		0.804 (0.778)
ln(MCap) (z)		-0.831 (1.57)		1.58 (1.02)
Deposit Beta (z)		-0.010 (0.816)		-0.169 (0.521)
N States Branches (z)		-2.72** (1.30)		-1.83** (0.719)
FOMC 2022 CAR (z)		-1.08 (0.806)		-1.39** (0.565)
Ret 2022-Q4 (z)		1.30 (0.969)		0.655 (0.584)
Ret 2022-Q3 (z)		-2.37** (1.13)		-0.396 (0.659)
Supply-driven Deposit Inflow 2020 (z)		1.46* (0.805)		1.11** (0.452)
Dep Inflow Uninsured 2022-Q4 (z)		-1.30 (1.76)		-1.70* (1.01)
Dep Inflow Insured 2022-Q4 (z)		0.165 (1.66)		-1.30 (1.07)
Constant	4.79*** (0.863)	4.46*** (1.06)	-0.915* (0.534)	-1.29** (0.569)
Observations	275	262	275	262
R ²	0.088	0.247	0.086	0.399
Adjusted R ²	0.064	0.171	0.062	0.339
Attention controls		✓		✓
Other valuation controls		✓		✓

This table presents cross-sectional OLS regression estimates relating quarterly bank deposit flows to Twitter pre-exposure, % Uninsured and % Loss MTM. The dependent variable, 'Deposit outflows (%)' is the percentage change in bank deposits from the end of Q4-2022 to the end of Q1-2023, multiplied by -1 so that a higher number indicates larger outflows. Columns 1 and 2 use the uninsured deposits (above the FDIC threshold of \$250K), columns 3 and 4 use all deposits to construct outflows. Columns 2 and 4 additionally include controls for bank valuation and outside attention, as well as controls for insured and uninsured deposit flows in Q4-2022, supply-driven deposit inflows in 2020, and measures of 2022 stock returns and the value of the deposit franchise, as in Table 4. Details on variable construction and data sources are provided in Appendix A.1. Robust standard errors are reported in parentheses.

* Indicate statistical significance at the 10% level.
** Indicate statistical significance at the 5% level.
*** Indicate statistical significance at the 1% level.

- **Total outflows.** The same effect is weaker for total deposits, which is intuitive because uninsured deposits are the most run-prone.
- **Punchline.** Social media does not just predict equity-market reactions; it is also linked to actual deposit flight.

5.5 Table 6 and Figure 8: hourly attention during the run

- **Read.** Table 6 estimates hourly return regressions with lagged Twitter attention.
- **Low run risk banks.** The interaction between post-March-9 and high lagged attention is small and insignificant.
- **High run risk banks.** The same interaction is negative and significant, around **-0.096** to **-0.100** basis points per hour in the full-window fixed-effects specifications.
- **Short window.** Around the immediate onset of the run, the magnitude becomes much larger, around **-0.147** to **-0.214**.
- **Figure 8.** Cumulative returns following high versus low Twitter attention diverge only in the high-run-risk subsample, and only once the run starts.

5.6 Table 7: retweets and alternative attention windows

- **Different lags.** The results are similar whether attention is measured over the past **1 hour**, **2 hours**, or **4 hours**. So the findings are not an artifact of a specific lag definition.

Table 6
Social media and the hunk run — hourly frequency[illegible]

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Table 6 (continued).
Panel D: Retaxnet (ETL)weighted Taxes

	Hourly and daily risk	
	(1)	(2)
1st Mar 00	-0.160**	-0.170**
2nd Mar 00	-0.000	-0.000
3rd Mar 00	-0.000	-0.000
4th Mar 00	-0.000	-0.000
5th Mar 00	-0.000	-0.000
6th Mar 00	-0.000	-0.000
7th Mar 00	-0.000	-0.000
8th Mar 00	-0.000	-0.000
9th Mar 00	-0.000	-0.000
10th Mar 00	-0.000	-0.000
11th Mar 00	-0.000	-0.000
12th Mar 00	-0.000	-0.000
13th Mar 00	-0.000	-0.000
14th Mar 00	-0.000	-0.000
15th Mar 00	-0.000	-0.000
16th Mar 00	-0.000	-0.000
17th Mar 00	-0.000	-0.000
18th Mar 00	-0.000	-0.000
19th Mar 00	-0.000	-0.000
20th Mar 00	-0.000	-0.000
21st Mar 00	-0.000	-0.000
22nd Mar 00	-0.000	-0.000
23rd Mar 00	-0.000	-0.000
24th Mar 00	-0.000	-0.000
25th Mar 00	-0.000	-0.000
26th Mar 00	-0.000	-0.000
27th Mar 00	-0.000	-0.000
28th Mar 00	-0.000	-0.000
29th Mar 00	-0.000	-0.000
30th Mar 00	-0.000	-0.000
31st Mar 00	-0.000	-0.000
1st Apr 00	-0.000	-0.000
2nd Apr 00	-0.000	-0.000
3rd Apr 00	-0.000	-0.000
4th Apr 00	-0.000	-0.000
5th Apr 00	-0.000	-0.000
6th Apr 00	-0.000	-0.000
7th Apr 00	-0.000	-0.000
8th Apr 00	-0.000	-0.000
9th Apr 00	-0.000	-0.000
10th Apr 00	-0.000	-0.000
11th Apr 00	-0.000	-0.000
12th Apr 00	-0.000	-0.000
13th Apr 00	-0.000	-0.000
14th Apr 00	-0.000	-0.000
15th Apr 00	-0.000	-0.000
16th Apr 00	-0.000	-0.000
17th Apr 00	-0.000	-0.000
18th Apr 00	-0.000	-0.000
19th Apr 00	-0.000	-0.000
20th Apr 00	-0.000	-0.000
21st Apr 00	-0.000	-0.000
22nd Apr 00	-0.000	-0.000
23rd Apr 00	-0.000	-0.000
24th Apr 00	-0.000	-0.000
25th Apr 00	-0.000	-0.000
26th Apr 00	-0.000	-0.000
27th Apr 00	-0.000	-0.000
28th Apr 00	-0.000	-0.000
29th Apr 00	-0.000	-0.000
30th Apr 00	-0.000	-0.000
1st May 00	-0.000	-0.000
2nd May 00	-0.000	-0.000
3rd May 00	-0.000	-0.000
4th May 00	-0.000	-0.000
5th May 00	-0.000	-0.000
6th May 00	-0.000	-0.000
7th May 00	-0.000	-0.000
8th May 00	-0.000	-0.000
9th May 00	-0.000	-0.000
10th May 00	-0.000	-0.000
11th May 00	-0.000	-0.000
12th May 00	-0.000	-0.000
13th May 00	-0.000	-0.000
14th May 00	-0.000	-0.000
15th May 00	-0.000	-0.000
16th May 00	-0.000	-0.000
17th May 00	-0.000	-0.000
18th May 00	-0.000	-0.000
19th May 00	-0.000	-0.000
20th May 00	-0.000	-0.000
21st May 00	-0.000	-0.000
22nd May 00	-0.000	-0.000
23rd May 00	-0.000	-0.000
24th May 00	-0.000	-0.000
25th May 00	-0.000	-0.000
26th May 00	-0.000	-0.000
27th May 00	-0.000	-0.000
28th May 00	-0.000	-0.000
29th May 00	-0.000	-0.000
30th May 00	-0.000	-0.000
31st May 00	-0.000	-0.000
1st Jun 00	-0.000	-0.000
2nd Jun 00	-0.000	-0.000
3rd Jun 00	-0.000	-0.000
4th Jun 00	-0.000	-0.000
5th Jun 00	-0.000	-0.000
6th Jun 00	-0.000	-0.000
7th Jun 00	-0.000	-0.000
8th Jun 00	-0.000	-0.000
9th Jun 00	-0.000	-0.000
10th Jun 00	-0.000	-0.000
11th Jun 00	-0.000	-0.000
12th Jun 00	-0.000	-0.000
13th Jun 00	-0.000	-0.000
14th Jun 00	-0.000	-0.000
15th Jun 00	-0.000	-0.000
16th Jun 00	-0.000	-0.000
17th Jun 00	-0.000	-0.000
18th Jun 00	-0.000	-0.000
19th Jun 00	-0.000	-0.000
20th Jun 00	-0.000	-0.000
21st Jun 00	-0.000	-0.000
22nd Jun 00	-0.000	-0.000
23rd Jun 00	-0.000	-0.000
24th Jun 00	-0.000	-0.000
25th Jun 00	-0.000	-0.000
26th Jun 00	-0.000	-0.000
27th Jun 00	-0.000	-0.000
28th Jun 00	-0.000	-0.000
29th Jun 00	-0.000	-0.000
30th Jun 00	-0.000	-0.000
1st Jul 00	-0.000	-0.000
2nd Jul 00	-0.000	-0.000
3rd Jul 00	-0.000	-0.000
4th Jul 00	-0.000	-0.000
5th Jul 00	-0.000	-0.000
6th Jul 00	-0.000	-0.000
7th Jul 00	-0.000	-0.000
8th Jul 00	-0.000	-0.000
9th Jul 00	-0.000	-0.000
10th Jul 00	-0.000	-0.000
11th Jul 00	-0.000	-0.000
12th Jul 00	-0.000	-0.000
13th Jul 00	-0.000	-0.000
14th Jul 00	-0.000	-0.000
15th Jul 00	-0.000	-0.000
16th Jul 00	-0.000	-0.000
17th Jul 00	-0.000	-0.000
18th Jul 00	-0.000	-0.000
19th Jul 00	-0.000	-0.000
20th Jul 00	-0.000	-0.000
21st Jul 00	-0.000	-0.000
22nd Jul 00	-0.000	-0.000
23rd Jul 00	-0.000	-0.000
24th Jul 00	-0.000	-0.000
25th Jul 00	-0.000	-0.000
26th Jul 00	-0.000	-0.000
27th Jul 00	-0.000	-0.000
28th Jul 00	-0.000	-0.000
29th Jul 00	-0.000	-0.000
30th Jul 00	-0.000	-0.000
31st Jul 00	-0.000	-0.000
1st Aug 00	-0.000	-0.000
2nd Aug 00	-0.000	-0.000
3rd Aug 00	-0.000	-0.000
4th Aug 00	-0.000	-0.000
5th Aug 00	-0.000	-0.000
6th Aug 00	-0.000	-0.000
7th Aug 00	-0.000	-0.000
8th Aug 00	-0.000	-0.000
9th Aug 00	-0.000	-0.000
10th Aug 00	-0.000	-0.000
11th Aug 00	-0.000	-0.000
12th Aug 00	-0.000	-0.000
13th Aug 00	-0.000	-0.000
14th Aug 00	-0.000	-0.000
15th Aug 00	-0.000	-0.000
16th Aug 00	-0.000	-0.000
17th Aug 00	-0.000	-0.000
18th Aug 00	-0.000	-0.000
19th Aug 00	-0.000	-0.000
20th Aug 00	-0.000	-0.000
21st Aug 00	-0.000	-0.000
22nd Aug 00	-0.000	-0.000
23rd Aug 00	-0.000	-0.000
24th Aug 00	-0.000	-0.000
25th Aug 00	-0.000	-0.000
26th Aug 00	-0.000	-0.000
27th Aug 00	-0.000	-0.000
28th Aug 00	-0.000	-0.000
29th Aug 00	-0.000	-0.000
30th Aug 00	-0.000	-0.000
31st Aug 00	-0.000	-0.000
1st Sep 00	-0.000	-0.000
2nd Sep 00	-0.000	-0.000
3rd Sep 00	-0.000	-0.000
4th Sep 00	-0.000	-0.000
5th Sep 00	-0.000	-0.000
6th Sep 00	-0.000	-0.000
7th Sep 00	-0.000	-0.000
8th Sep 00	-0.000	-0.000
9th Sep 00	-0.000	-0.000
10th Sep 00	-0.000	-0.000
11th Sep 00	-0.000	-0.000
12th Sep 00	-0.000	-0.000
13th Sep 00	-0.000	-0.000
14th Sep 00	-0.000	-0.000
15th Sep 00	-0.000	-0.000
16th Sep 00	-0.000	-0.000
17th Sep 00	-0.000	-0.000
18th Sep 00	-0.000	-0.000
19th Sep 00	-0.000	-0.000
20th Sep 00	-0.000	-0.000
21st Sep 00	-0.000	-0.000
22nd Sep 00	-0.000	-0.000
23rd Sep 00	-0.000	-0.000
24th Sep 00	-0.000	-0.000
25th Sep 00	-0.000	-0.000
26th Sep 00	-0.000	-0.000
27th Sep 00	-0.000	-0.000
28th Sep 00	-0.000	-0.000
29th Sep 00	-0.000	-0.000
30th Sep 00	-0.000	-0.000
1st Oct 00	-0.000	-0.000
2nd Oct 00	-0.000	-0.000
3rd Oct 00	-0.000	-0.000
4th Oct 00	-0.000	-0.000
5th Oct 00	-0.000	-0.000
6th Oct 00	-0.000	-0.000
7th Oct 00	-0.000	-0.000
8th Oct 00	-0.000	-0.000
9th Oct 00	-0.000	-0.000
10th Oct 00	-0.000	-0.000
11th Oct 00	-0.000	-0.000
12th Oct 00	-0.000	-0.000
13th Oct 00	-0.000	-0.000
14th Oct 00	-0.000	-0.000
15th Oct 00	-0.000	-0.000
16th Oct 00	-0.000	-0.000
17th Oct 00	-0.000	-0.000
18th Oct 00	-0.000	-0.000
19th Oct 00	-0.000	-0.000
20th Oct 00	-0.000	-0.000
21st Oct 00	-0.000	-0.000
22nd Oct 00	-0.000	-0.000
23rd Oct 00	-0.000	-0.000
24th Oct 00	-0.000	-0.000
25th Oct 00	-0.000	-0.000
26th Oct 00	-0.000	-0.000
27th Oct 00	-0.000	-0.000
28th Oct 00	-0.000	-0.000
29th Oct 00	-0.000	-0.000
30th Oct 00	-0.000	-0.000
31st Oct 00	-0.000	-0.000
1st Nov 00	-0.000	-0.000
2nd Nov 00	-0.000	-0.000
3rd Nov 00	-0.000	-0.000
4th Nov 00	-0.000	-0.000
5th Nov 00	-0.000	-0.000
6th Nov 00	-0.000	-0.000
7th Nov 00	-0.000	-0.000
8th Nov 00	-0.000	-0.000
9th Nov 00	-0.000	-0.000
10th Nov 00	-0.000	-0.000
11th Nov 00	-0.000	-0.000
12th Nov 00	-0.000	-0.000
13th Nov 00	-0.000	-0.000
14th Nov 00	-0.000	-0.000
15th Nov 00	-0.000	-0.000
16th Nov 00	-0.000	-0.000
17th Nov 00	-0.000	-0.000
18th Nov 00	-0.000	-0.000
19th Nov 00	-0.000	-0.000
20th Nov 00	-0.000	-0.000
21st Nov 00	-0.000	-0.000
22nd Nov 00	-0.000	-0.000
23rd Nov 00	-0.000	-0.000
24th Nov 00	-0.000	-0.000
25th Nov 00	-0.000	-0.000
26th Nov 00	-0.000	-0.000
27th Nov 00	-0.000	-0.000
28th Nov 00	-0.000	-0.000
29th Nov 00	-0.000	-0.000
30th Nov 00	-0.000	-0.000
1st Dec 00	-0.000	-0.000
2nd Dec 00	-0.000	-0.000
3rd Dec 00	-0.000	-0.000
4th Dec 00	-0.000	-0.000
5th Dec 00	-0.000	-0.000
6th Dec 00	-0.000	-0.000
7th Dec 00	-0.000	-0.000
8th Dec 00	-0.000	-0.000
9th Dec 00	-0.000	-0.000
10th Dec 00	-0.000	-0.000
11th Dec 00	-0.000	-0.000
12th Dec 00	-0.000	-0.000
13th Dec 00	-0.000	-0.000
14th Dec 00	-0.000	-0.000
15th Dec 00	-0.000	-0.000
16th Dec 00	-0.000	-0.000
17th Dec 00	-0.000	-0.000
18th Dec 00	-0.000	-0.000
19th Dec 00	-0.000	-0.000
20th Dec 00	-0.000	-0.000
21st Dec 00	-0.000	-0.000
22nd Dec 00	-0.000	-0.000
23rd Dec 00	-0.000	-0.000
24th Dec 00	-0.000	-0.000
25th Dec 00	-0.000	-0.000
26th Dec 00	-0.000	-0.000
27th Dec 00	-0.000	-0.000
28th Dec 00	-0.000	-0.000
29th Dec 00	-0.000	-0.000
30th Dec 00	-0.000	-0.000
31st Dec 00	-0.000	-0.000
1st Jan 01	-0.000	-0.000
2nd Jan 01	-0.000	-0.000
3rd Jan 01	-0.000	-0.000
4th Jan 01	-0.000	-0.000
5th Jan 01	-0.000	-0.000
6th Jan 01	-0.000	-0.000
7th Jan 01	-0.000	-0.000
8th Jan 01	-0.000	-0.000
9th Jan 01	-0.000	-0.000
10th Jan 01	-0.000	-0.000
11th Jan 01	-0.000	-0.000
12th Jan 01	-0.000	-0.000
13th Jan 01	-0.000	-0.000
14th Jan 01	-0.000	-0.000
15th Jan 01	-0.000	-0.000
16th Jan 01	-0.000	-0.000
17th Jan 01	-0.000	-0.000
18th Jan 01	-0.000	-0.000
19th Jan 01	-0.000	-0.000
20th Jan 01	-0.000	-0.000
21st Jan 01	-0.000	-0.000</

*** Indicate statistical significance at the 1% level.

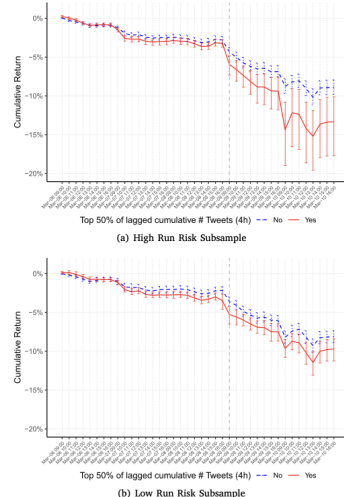


Fig. 8. Cumulative Returns Following High versus Low Twitter Attention. This figure shows the average cumulative returns (publicly listed banks and bank holding companies (BHCs)) from March 6 until March 10, 2023 (during the trading hours), splitting the sample into banks with high run risk (Panel (a)) and low run risk (Panel (b)). In each subsample we split the sample of bank stocks into those with above (red solid line) and below (blue dashed line) median number of tweets over the previous four hours, calculate average returns for the two groups, and cumulate the average returns at the hourly frequency from March 6 until March 10, 2023. The gray vertical dashed line indicates March 09, 2023 at 9am. 95% confidence intervals around the mean cumulative return are constructed by bootstrapping with clustering at the bank level.

- **Retweet heterogeneity.** The attention effect is much stronger when recent tweets receive many retweets. In the fixed-effects specifications, the key interaction is about **-0.124** in the high-retweet subsample versus about **-0.075** in the low-retweet subsample.
- **Punchline.** What matters is not just raw tweet volume, but socially propagated tweet volume.

Table 7
Hourly frequency — different lags and tweet attention

Panel A: Baseline specification with different lags			
	Hourly stock return		
	(1)	(2)	
1/4° Tweeks (H) 0 - 1	0.066 ^{***} (0.018)	0.059 ^{***} (0.015)	4 h
1/4° Tweeks (H) 2 - 3	-0.024 ^{***} (0.008)	-0.024 ^{***} (0.008)	4 h
Observations	14,566	10,946	14,566
Walds χ^2	0.244	0.244	0.245
Bayes Criteria - Information	0.003	0.003	0.003
Bayes Criteria - Information	✓	✓	✓
Fit by FE	✓	✓	✓
Panel B: Tweeks with high- vs low number of mentions	Hourly stock return		
	(1)	(2)	
1/4° Tweeks (H) 0 - 1	-0.226 ^{***} (0.045)	-0.228 ^{***} (0.046)	4 h
1/4° Tweeks (H) 2 - 3	0.065 ^{***} (0.012)	0.061 ^{***} (0.012)	4 h
1/4° Tweeks (H) 0 - 1 & H 2 - 3	-0.087 ^{***} (0.006)	-0.079 ^{***} (0.006)	4 h
Observations	14,566	10,946	14,566
Walds χ^2	0.937	0.952	0.931
Bayes Criteria - Information	0.006	0.006	0.006
Bayes Criteria - Information	✓	✓	✓
Fit by FE	✓	✓	✓

This table presents OLS regressions that relate hourly stock returns to lags in high- versus low-mention tweeks, estimated as follows: $RET_{it} = \alpha + \beta_1 Tweeks_{it}^{H(0-1)} + \beta_2 Tweeks_{it}^{H(2-3)} + \beta_3 Tweeks_{it}^{H(0-1) \& H(2-3)} + \epsilon_{it}$, where RET_{it} is the hourly stock return on the pretent 1, 2, 3, and 4 h, respectively, to construct 1/4° tweeks (H) 0 - 1, 2 - 3, and 0 - 1 & 2 - 3. Panel A reports the baseline specification, and Panel B reports the specification with high- and low-mention tweeks. The results are reported for the two subsamples: stock returns and overnight returns and their interaction with 1/4° tweeks as well as on day and day after the event. The results are calculated based on 14,566 observations reported in specifications are clustered at the hour level.

***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

** Indicate statistical significance at the 5% level.

Table 8
High frequency return responses to tweet sentiment

[illegible]

*** Indicate statistical significance at the 1% level.

Table 9
High frequency tests, Controlling for networks and followers

[illegible]

¹²² Indicates statistical significance at the 5% level.

5.7 Table 8 and Table 9: tweet-level market impacts

- **Read.** These tables ask which kinds of tweets move prices in the minutes right after they are posted.
- **Tech-community tweets.** Negative sentiment in tech-community tweets has a strong immediate effect, around **-11 to -12 basis points**.

- **Run tweets.** Negative sentiment in run tweets has an even stronger effect, around **-13 basis points**.
- **Contagion tweets.** Negative sentiment in contagion tweets has the largest effect, around **-20 to -23 basis points**.
- **High run risk interaction.** The interaction between negative sentiment and high run risk is small and insignificant.
- **Table 9.** Adding controls for follower counts and retweets does not materially change the main interaction estimates.
- **Punchline.** Run-relevant communication moves prices immediately, but negative sentiment per se does *not* explain why some banks are more run-prone than others. That role belongs to attention and coordination.

6 Short summary

- **Method.** Measure ex ante Twitter exposure before the SVB collapse, then relate it to run-period stock losses, deposit outflows, and high-frequency return reactions.
- **Identification.** Use pre-run attention, weak correlation between pre-exposure and run risk, rich controls, lagged hourly attention, and minute-level event windows around tweets.
- **Application.** U.S. publicly traded banks and bank holding companies during the March 2023 banking stress episode.
- **Main result.** Social media **amplifies classical bank run risks**. Attention and run-relevant communication matter; negative sentiment alone does not.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper shows that banks with greater pre-existing exposure to Twitter experienced a more severe run in March 2023.
- This was especially true for banks with the classic fragilities of **high uninsured deposits** and **large mark-to-market losses**.
- The evidence is strongest for **Twitter attention** and the social propagation of run-relevant messages, not for sentiment-driven pessimism alone.

7.2 Economic intuition

- In a fragile banking system, uninsured depositors must coordinate to run.
- Social media lowers the cost and increases the speed of that coordination.

- A bank that is already salient on social media becomes easier to focalize during stress, so once bad news hits, attention can quickly turn into pressure.
- The key mechanism is therefore not that Twitter reveals hidden insolvency with great precision, but that it helps convert existing fragility into realized distress.

7.3 Takeaway

This paper's main lesson is that social media should be treated as part of the modern bank-run technology. Traditional balance-sheet weaknesses still determine which banks are fragile, but social media can accelerate and magnify the pressure on those banks by concentrating depositor attention and communication exactly when coordination matters most.